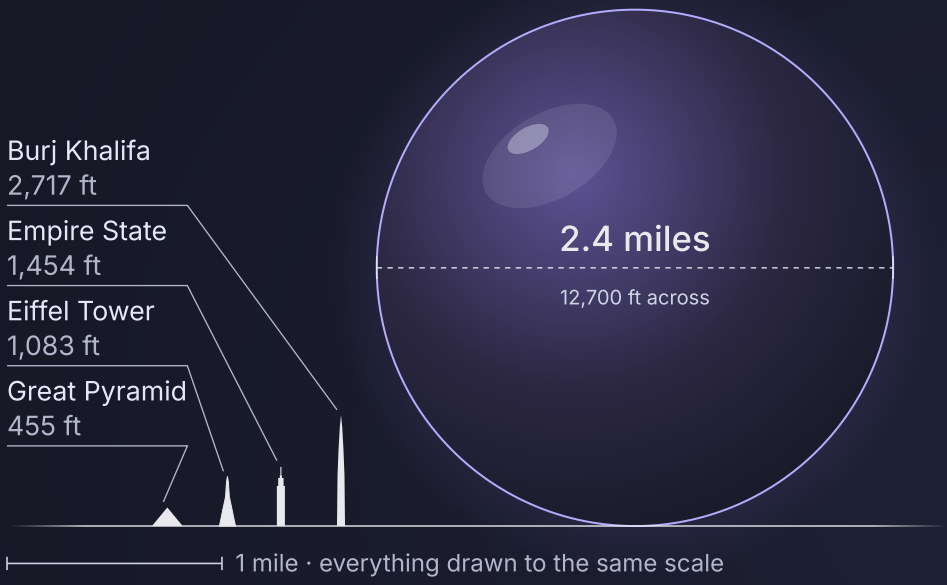
KANSAS · SOIL MOISTURE

KANSAS STATE UNIVERSITY | Agronomy
Soil Water Processes Lab

7.3 mi³

The most plant available water the top 24 inches (2 feet) of Kansas soils can hold, with the whole state at field capacity.

≈ 8.0 trillion gallons · ≈ 25 million acre-feet



Gathered into one marble, it would stand **more than 4½ times taller** than the world's tallest building.

12 million

Olympic swimming pools — one pool = 660,000 gal

94%

of full Lake Mead, the largest man-made reservoir in the U.S.

5.6 in

of a single rainfall event soaking the entire state at once

CLOSER TO HOME

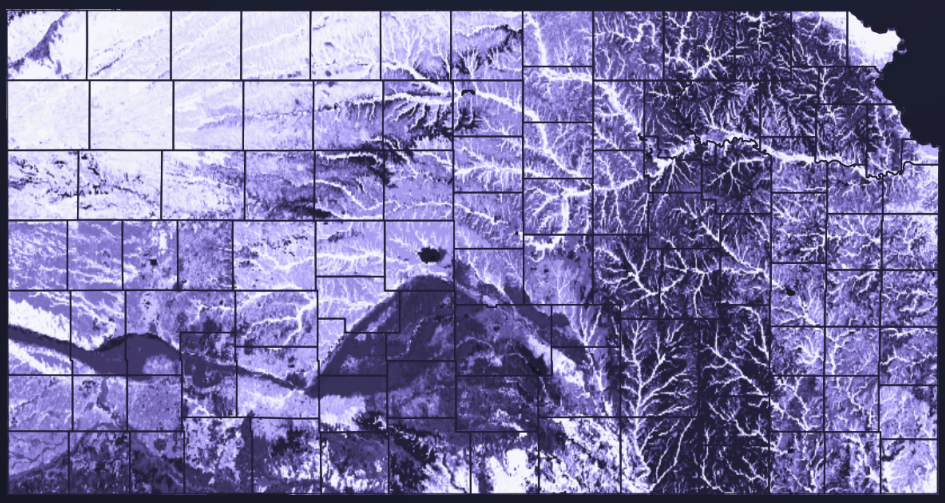
5½ times

That's how many times this water could fill every lake and pond in Kansas combined (≈ 4.4 million acre-ft).

Or, the ten largest lakes one at a time:

Milford	366,000 acre-ft	67×
Wilson	237,000 acre-ft	103×
Tuttle Creek	232,000 acre-ft	106×
Waconda	209,000 acre-ft	117×
Perry	193,000 acre-ft	127×
Cedar Bluff	169,000 acre-ft	145×
Cheney	156,000 acre-ft	157×
El Dorado	156,000 acre-ft	157×
Melvern	148,000 acre-ft	166×
Clinton	116,000 acre-ft	211×

WHERE IT IS STORED



≤ 4.9 in ≥ 6.2 in

Silt loams in the northwest hold the most plant available water; heavy clay soils in the east and sandy soils along the Arkansas River hold the least.

WHY IT MATTERS

Crops, pastures, and native grasslands depend on this stored soil water to sustain transpiration and produce grain and forage between rains. How full the soil profile is at any moment, known as the fraction of available water, is a robust indicator of agricultural drought.

Plant available water is what a soil holds after excess water drains away (field capacity) minus what it holds too tightly for roots to extract (permanent wilting point). We report the top 24 inches (2 feet), the most dynamic and densely rooted part of the soil profile, where water content increases within hours of rainfall and declines over the following days through soil evaporation and plant uptake. Both limits were estimated with custom pedotransfer functions developed by the K-State Soil Water Processes Lab, using clay, sand, bulk density, and organic matter maps from the USDA Natural Resources Conservation Service. Figures rounded: full Lake Mead ≈ 26 million acre-ft; Kansas area ≈ 82,300 mi²; lakes and ponds at normal pool (Rahmani et al. 2018; USACE National Inventory of Dams; Callihan 2013). Created by Andres Patrignani, Soil Water Processes Lab, Agronomy, Kansas State University, October 2026.